

3)

A) CODE OUTPUT:

```
nithya@WIN-482M0QMA856:~$ cc practical3.c
nithya@WIN-482M0QMA856:~$ ./a.out
Before fork
PID: 2762
PPID: 332

Parent process
PID: 2762
PPID: 332

Parent process is running
Child process
PID: 2763
PPID: 2762
Child process is running
```

B) Commands

ps:

```
nithya@WIN-482M0QMA856:~$ ps
  PID TTY          TIME CMD
   332 pts/0        00:00:00 bash
  2820 pts/0        00:00:00 ps
```

Experiment:

```

nithya@WIN-482M0QMA856:~$ sleep 100 &
[2] 2548
nithya@WIN-482M0QMA856:~$ ps -p 2548 -o pid,stat,cmd
  PID STAT CMD
  2548 S    sleep 100
nithya@WIN-482M0QMA856:~$ cat /proc/2548/status
Name:      sleep
Umask:     0022
State:     S (sleeping)
Tgid:      2548
Ngid:      0
Pid:       2548
PPid:      364
TracerPid: 0
Uid:       1000    1000    1000    1000
Gid:       1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 100 1000
NSTgid:    2548
NSpid:     2548
NSpgid:    2548
NSsid:     364
Kthread:   0
VmPeak:    16112 kB
VmSize:    16112 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     7784 kB
VmRSS:     7784 kB
RssAnon:           1224 kB
RssFile:           6560 kB
RssShmem:          0 kB
VmData:      288 kB
VmStk:       132 kB
VmExe:      4832 kB
VmLib:      3240 kB
VmPTE:        72 kB
VmSwap:      0 kB
HugetlbPages:      0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:     1

```

```

Mems_allowed_list:      0
voluntary_ctxt_switches: 1
nonvoluntary_ctxt_switches: 0
nithya@WIN-482M0QMA856:~$ kill 2548
[2]+  Terminated                  sleep 100
nithya@WIN-482M0QMA856:~$ |

```

top:

```
nithya@WIN-482M0QMA856:~$ top
top - 17:19:07 up 45 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 34 total, 1 running, 33 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 6849.7 total, 6001.8 free, 610.9 used, 386.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 6238.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24568	15500	11560	S	0.0	0.2	0:01.03	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
49	root	19	-1	50428	16824	15500	S	0.0	0.2	0:00.42	systemd-journal
66	systemd+	20	0	22416	14524	12032	S	0.0	0.2	0:00.13	systemd-resolve
88	root	20	0	35252	12380	9256	S	0.0	0.2	0:02.32	systemd-udev
140	root	20	0	2888	2008	1880	S	0.0	0.0	0:00.07	chronyd-starter
145	root	20	0	4472	3156	2908	S	0.0	0.0	0:00.01	cron
148	message+	20	0	8820	5404	4724	S	0.0	0.1	0:00.09	dbus-daemon
168	root	20	0	44100	29712	14568	S	0.0	0.4	0:00.29	networkd-dispat
176	root	20	0	18436	9412	8172	S	0.0	0.1	0:00.15	systemd-logind
201	syslog	20	0	220548	5828	4580	S	0.0	0.1	0:00.11	rsyslogd
271	_chrony	20	0	23592	11216	7260	S	0.0	0.2	0:00.11	chronyd
274	_chrony	20	0	12096	2436	1792	S	0.0	0.0	0:00.00	chronyd
289	root	20	0	123464	32788	18180	S	0.0	0.5	0:00.21	unattended-upgr
292	root	20	0	5492	2724	2540	S	0.0	0.0	0:00.00	agetty
330	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
331	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.08	Relay(332)
332	nithya	20	0	6268	5700	3992	S	0.0	0.1	0:00.21	bash
333	root	20	0	8644	5480	4640	S	0.0	0.1	0:00.02	login
380	nithya	20	0	22760	13408	10940	S	0.0	0.2	0:00.27	systemd
382	nithya	20	0	22916	4116	2116	S	0.0	0.1	0:00.00	(sd-pam)
408	nithya	20	0	6268	5612	3992	S	0.0	0.1	0:00.04	bash
736	nithya	20	0	8708	5300	4748	S	0.0	0.1	0:00.08	dbus-daemon
740	nithya	20	0	380960	8092	7440	S	0.0	0.1	0:00.02	at-spi-bus-laun
747	nithya	20	0	8544	5096	4724	S	0.0	0.1	0:00.02	dbus-daemon
748	nithya	20	0	476880	17808	14908	S	0.0	0.3	0:00.13	xdg-desktop-por
753	nithya	20	0	304804	6720	6220	S	0.0	0.1	0:00.02	xdg-permission-
758	nithya	20	0	618996	7620	6940	S	0.0	0.1	0:00.03	xdg-document-po
764	root	20	0	2792	2308	2192	S	0.0	0.0	0:00.00	fusermount3
767	nithya	20	0	482104	23840	18892	S	0.0	0.3	0:00.17	xdg-desktop-por
776	nithya	20	0	168688	7832	7180	S	0.0	0.1	0:00.03	at-spi2-registr

```

hithya@WIN-482M0QMA856:~$ uname
Linux
hithya@WIN-482M0QMA856:~$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            48 bits physical, 48 bits virtual
Byte Order:               Little Endian
CPU(s):                   16
On-line CPU(s) list:      0-15
Vendor ID:                AuthenticAMD
Model name:               AMD Ryzen 7 7730U with Radeon Graphics
CPU family:               25
Model:                    80
Thread(s) per core:       2
Core(s) per socket:       8
Socket(s):                 1
Stepping:                  0
BogoMIPS:                  3992.39
Flags:                     fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall
nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep
_good nopl xtopology tsc_reliable nonstop_tsc cpuid ex
td_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand h
ypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a
misalignsse 3dnowprefetch osvw topoext perfctr_core ss
bd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb sha_ni
xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat
npt nrip_save tsc_scale vmcb_clean flushbyasid decodea
ssists pausefilter pfthreshold v_vmsave_vmload umip va
es vpcmlulqdq rdpid fsrm
Virtualization features:
Virtualization:           AMD-V
Hypervisor vendor:        Microsoft
Virtualization type:       full
Caches (sum of all):
L1d:                       256 KiB (8 instances)
L1i:                       256 KiB (8 instances)
L2:                         4 MiB (8 instances)
L3:                        16 MiB (1 instance)
NUMA:
NUMA node(s):              1
NUMA node0 CPU(s):         0-15

```

```

nithya@WIN-482M0QMA856:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro
/

nithya@WIN-482M0QMA856:~$ ps
  PID TTY          TIME CMD
  364 pts/0    00:00:00 bash
 1980 pts/0    00:00:00 ps

nithya@WIN-482M0QMA856:~$ top
top - 13:33:22 up 59 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  25 total,   1 running, 24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  6849.6 total,  6095.2 free,   609.1 used,   299.6 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  6240.5 avail Mem

   PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24284   15372  11516 S   0.0   0.2   0:01.89 systemd
    2 root        20   0    3180    2212   2072 S   0.0   0.0   0:00.02 init-syste+
    6 root        20   0    3180    2040   1940 S   0.0   0.0   0:00.00 init
   52 root        19  -1   50400   16764  15464 S   0.0   0.2   0:00.54 systemd-jo+
   69 systemd+    20   0   22416   14444  11944 S   0.0   0.2   0:00.18 systemd-re+
   91 root        20   0   35260   12288   9176 S   0.0   0.2   0:03.17 systemd-ud+

```

Command

Role/Purpose

uname	Displays the os being used.
lscpu	Displays CPU architecture, cores, threads, and processor information.
lsblk	Displays storage devices, partitions, and their sizes.
ps	Displays currently running processes and their PIDs.
top	Provides real-time information about processes and CPU/memory usage.